

Ex 2.1.5 Show for a sphere of radius  $a$  parametrized w/  $u = \theta, v = \phi$  that  $(g_{ab}) = \begin{pmatrix} a^2 & 0 \\ 0 & a^2 \sin^2 \theta \end{pmatrix}$  and only non-zero connection coefficients are  $\Gamma_{22}^1 = -\sin \theta \cos \theta$  and  $\Gamma_{12}^2 = \Gamma_{21}^2 = \cot \theta$ .

From Example (P.8 of Notes):

From Example 1.1.4, we get  $e_1 = e_\theta, e_2 = e_\phi$ , and then  $g_{11} = e_1 \cdot e_1$ , etc.

$$g_{11} = a^2 \quad g_{22} = a^2 \sin^2 \theta \quad \text{and} \quad g_{12} = g_{21} = 0$$

$$g^{11} = \frac{1}{a^2} \quad g^{22} = \frac{1}{a^2 \sin^2 \theta} \quad \text{and} \quad g^{12} = g^{21} = 0$$

The connection coefficients are  $\Gamma_{bc}^a = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc})$

The only non-zero  $\partial_d g_{bc}$  term is  $\partial_1 g_{22} = 2a^2 \sin \theta \cos \theta$ :

$$g_{12} = g_{21} = 0 \quad \text{and} \quad \partial_\theta g_{11} = \partial_\phi g_{11} = 0 \quad \text{and} \quad \partial_2 g_{22} = \frac{\partial}{\partial \phi} (a^2 \sin^2 \theta) = 0$$

Consider  $\Gamma_{bc}^1$ :  $\frac{1}{2} g^{1d} \Rightarrow d=1$  to get  $\frac{1}{2} g^{11}$ . So  $g_{1c} = g_{b1} = 0$ , leaving  $\partial_d g_{bc} = \partial_1 g_{bc}$   
 $\Rightarrow b=c=2$

$$\text{So } \Gamma_{22}^1 = \frac{1}{2} g^{1d} (-\partial_d g_{bc}) = -\frac{1}{2} g^{11} (\partial_1 g_{22}) = -\frac{2a^2 \sin \theta \cos \theta}{2a^2} = -\sin \theta \cos \theta \quad \checkmark$$

Consider  $\Gamma_{bc}^2$ :  $\frac{1}{2} g^{2d} \Rightarrow d=2$  to get  $\frac{1}{2} g^{22}$ . So  $\Gamma_{bc}^2 = \frac{1}{2} g^{22} (\partial_b g_{ac} + \partial_c g_{b2} - \partial_2 g_{bc})$

So to be non-zero,  $\partial_b g_{ac} = \partial_1 g_{22}$ ;  $\partial_c g_{b2} = \partial_1 g_{22}$ ; but  $\partial_2 g_{bc} = 0$   
 $b=1 \neq c=2 \quad c=1 \text{ and } b=2$

$$\therefore \Gamma_{12}^2 = \Gamma_{21}^2 = \frac{1}{2} g^{22} \partial_1 g_{22} = \frac{2a^2 \sin \theta \cos \theta}{2a^2 \sin^2 \theta} = \cot \theta \quad \checkmark$$

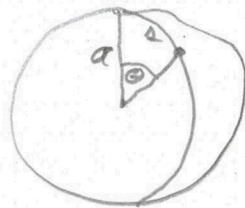
Exercise 2.1.6 Show that all longitude lines on a sphere are geodesics.

A line of longitude runs from the north pole to the south pole. Such a line has eq  $\phi = \phi_0$ .

It is parametrized by  $u^1 = u^\theta = \theta = \frac{s}{a}$  where  $r = a$ , and  $u^2 = u^\phi = \phi_0$

We can write this as  $u^A = \frac{s}{a} \delta_1^A + \phi_0 \delta_2^A$  for  $A=1, 2$ .

Thus  $\dot{u}^A \equiv \frac{du^A}{ds} = \frac{1}{a} \delta_1^A$  and  $\ddot{u}^A = 0$ . The geodesic equations are



$$\ddot{u}^A + \Gamma_{BC}^A \dot{u}^B \dot{u}^C = \frac{1}{a^2} \delta_1^B \delta_1^C \Gamma_{BC}^A \stackrel{(2.11)}{=} 0 \quad \text{for } A=1, 2 \Leftrightarrow \delta_1^B \delta_1^C \Gamma_{BC}^A = 0$$

From Exercise 2.1.5, the only non-zero connection coefficients are  $\Gamma_{22}^1 = -\sin \theta \cos \theta$  and  $\Gamma_{12}^2 = \Gamma_{21}^2 = \cot \theta$

So, for  $A=1$  the only possible non-zero term is  $\delta_1^B \delta_1^C \Gamma_{BC}^1 = \delta_1^2 \delta_1^2 \Gamma_{22}^1 = 0$  and for  $A=2$

$$\delta_1^B \delta_1^C \Gamma_{BC}^2 = \delta_1^2 \delta_1^1 \Gamma_{21}^2 + \delta_1^1 \delta_1^2 \Gamma_{12}^2 = 0. \quad \text{Since both eqs are always true, the longitude}$$

line  $\phi = \phi_0$  is a geodesic.